Submission Summary

🔹 **Task 1 – Long Form ASR Transcription**

1. Overview of the Final Model: Our final Automatic Speech Recognition (ASR) system is based on Hugging Face Transformers implementation of OpenAI Whisper. We initialized from the pretrained checkpoint tugstugi/bengaliai-regional-asr_whisper-medium (769M parameters) and further fine-tuned the model exclusively on the competition dataset to better adapt it to the domain characteristics and long-form speech patterns.

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2. Model Parameter Size

- Base Model: tugstugi/bengaliai-regional-asr_whisper-medium
- Total Parameters: ~769 Million

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3. Inference Time: Inference was performed on a single T4 GPU (Kaggle environment).

- Total inference time (full test set): ~**3 hours 38 minutes [submitted Inference Notebook]**
- Average per audio file (default settings): **8-9 minutes**
- Total inference time (full test set): ~ **25 minute 36 seconds [Super Faster Inference,** WER deviates by 1 point only but still in **#5** in public and private leaderboard**]**
- Optimized decoding configuration: *****1 minute*** per audio Only**

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4. Model Artifact

The fine-tuned model checkpoint is publicly available on Hugging Face:

🔗 **https://huggingface.co/bitwisemind/sam_15000_clean_text_full_model**

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5. Datasets Used: Only the official competition dataset was used for training and validation.

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6. Training Details: bengaliAI/tugstugi_bengaliai-regional-asr_whisper-medium | 90/10 split (1–15229 segments of whole data), batch 8×2=16, AdamW 8bit (5e-6), cosine, 6 epochs, fp16 + grad checkpointing, eval/save every 6000 (limit 2). Max 30s audio / 1000 text, augmentation 0.3, gen max 225

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7. Training / Fine-Tuning/ Inference Notebooks

- **GitHub:**https://github.com/sazzadadib/BitwiseMind_DL_Sprint_4.0/tree/main/Bengali%20Long-form%20Speech%20Recognition

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🔹 **Task 2 – Speaker Diarization**

1. Overview of the Final Model: Our speaker diarization system is built using the pyannote.ai diarization pipeline, powered by the pyannote.audio framework. We used the pretrained segmentation model:**pyannote/segmentation-3.0**. This model detects speaker change boundaries and generates time-stamped speaker segments. The diarization pipeline then clusters segments into distinct speaker identities.

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2. Model Parameter Size

- **Segmentation Model: pyannote/segmentation-3.0**
- **Total Parameters: ~1.5 Million**

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3. Inference Time: Inference was performed on a single T4 GPU (Kaggle environment).

- Total inference time (full test set): ~1 hour 20 minutes
- Average per audio file: 5–6 minutes

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4. Model Artifact

The trained diarization checkpoint is available on Hugging Face:

🔗 Finetuned Segmentation Model:https://huggingface.co/datasets/Amanafif554/diarizaaaaaation_CKPT

segmentation-epoch=39.ckpt, was used in inference pipeline.

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5. Datasets Used: Only the official competition dataset was used.

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6. Training Details: pyannote/segmentation-3.0 with pyannote/speaker-diarization-community-1 | 10s chunks, max 3 speakers/chunk (1/frame), 50 epochs on 1 GPU (Lightning). Monitor loss/train (min), save best+last, metric DER, protocol CustomData.SpeakerDiarization.train with train/dev/test splits.

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7. Training / Fine-Tuning/ Inference Notebooks

- **GitHub:**https://github.com/sazzadadib/BitwiseMind_DL_Sprint_4.0/tree/main/Bengali%20Speaker%20Diarization